iWaaave Pro SE

Digital Audio Visualizer

User Manual · Rev 2.0 · DE1-SoC FPGA Platform · VGA Extension

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Field	Value
Platform	Terasic DE1-SoC (Cyclone V FPGA)
System Clock	50 MHz (CLOCK_50)
Audio Sample Rate	48 kHz, 24-bit stereo (Wolfson WM8731)
Display	16×16 LED Matrix (Red + Green channels), GPIO_1
Modes	Amplitude Time-Domain (Green) / FFT Frequency-Domain (Red)
Audio Input	3.5mm Mic-In Port (AUD_ADCDAT, AUD_BCLK, AUD_ADCLRCK)
VGA Output	External monitor via 15-pin D-Sub · 640×480 @ 25.175 MHz
User Controls	SW[9]: Power · SW[0]: Viz Mode · SW[7]: VGA On/Off · SW[6]: BG Color
Debug Output	LEDR[9:0] on-board LEDs (amplitude monitor)

1. System Overview

iWaaave Pro SE is a real-time audio visualizer implemented on the Terasic DE1-SoC FPGA development board. The system captures audio from a standard 3.5mm Mic-In connection, processes it digitally in hardware, and renders the result simultaneously on an attached 16×16 dual-color LED matrix and an external VGA monitor.

Two visualization modes are available, selectable in real time using the SW[0] switch:

- Amplitude Mode (SW[0] = 0) — displays a scrolling time-domain waveform using the Green LED channel. Each column represents the averaged absolute amplitude of 1024 consecutive audio samples, giving a smooth, rolling oscilloscope-like display at approximately 46 frames per second.
- FFT Frequency Mode (SW[0] = 1) — displays a live frequency-domain spectrum using the Red LED channel. A 16-point Fast Fourier Transform (Cooley-Tukey algorithm) decomposes incoming audio into 16 frequency bins, with each bin driving one column of the LED matrix. Bin height reflects signal magnitude on a logarithmic scale.

Both visualization modes are rendered on two independent output systems simultaneously. The 16×16 LED matrix (GPIO_1) shows a compact single-color bar graph, while the VGA display (controlled by SW[7] and SW[6]) shows a full-color dual-panel view with the Amplitude waveform on the top half and the FFT spectrum on the bottom half.

The system performs simultaneous audio passthrough: audio received on the Mic-In port is also forwarded to the DAC output in real time with no added latency, allowing the board to be used as an inline monitoring device.

2. Hardware Setup

2.1 Required Hardware

- Terasic DE1-SoC FPGA development board
- 16×16 dual-color LED matrix panel (P10 format, Red + Green, connected to GPIO_1)
- 3.5mm stereo audio cable (Tip-Ring-Sleeve)
- Audio source: microphone, headphone output, instrument, or any line-level device
- VGA monitor with a 15-pin D-Sub connector (optional, for VGA display feature)
- VGA cable (15-pin D-Sub male to monitor input)
- USB-Blaster cable for programming (if applicable)

2.2 Audio Connection

iWaaave Pro SE receives audio through the DE1-SoC Mic-In port, which is a standard 3.5mm TRS jack on the board. Connect your audio source as follows:

1. Locate the 3.5mm Mic-In jack on the DE1-SoC board (labeled MIC on the audio codec section).
2. Insert a standard 3.5mm audio cable into the Mic-In port.

3. Connect the other end to your audio source — for example, the headphone output of a phone, laptop, or audio interface.
4. Ensure the source is playing audio or generating a signal before observing the display.

NOTE	The Mic-In port accepts both microphone-level and line-level signals. For best display results, use a line-level source (such as a headphone jack) at moderate volume. Very loud signals may saturate the ADC and clip the visualization.

NOTE	The system simultaneously outputs the same audio signal on the DAC (line-out) port. This passthrough is handled by the always_ff DAC Passthrough block, which forwards adc_left and adc_right to dac_left and dac_right on every audio_advance pulse.

2.3 LED Matrix Connection

Connect the 16×16 LED matrix to the GPIO_1 header (36 pins) on the DE1-SoC board. The LEDDriver module drives the matrix with a scan clock of approximately 1526 Hz derived by dividing CLOCK_50 by 2^15.

- Red pixels are active in FFT Frequency Mode (SW[0] = 1).
- Green pixels are active in Amplitude Mode (SW[0] = 0).
- Both channels can theoretically be active simultaneously in extended configurations, producing yellow where both R and G are set.

2.4 VGA Monitor Connection

iWaaave Pro SE outputs a standard VGA video signal from the DE1-SoC board's built-in VGA DAC. To use the VGA display feature:

5. Connect a standard 15-pin D-Sub VGA cable from the DE1-SoC VGA connector to your monitor.
6. Set your monitor to its VGA input if it has multiple input sources.
7. Once the system is powered on (SW[9] = 1), flip SW[7] UP to enable the VGA output.
8. The monitor will display a dual-panel view: Amplitude waveform on the top half, FFT spectrum on the bottom half.

NOTE	The VGA output continues to generate valid sync signals even when SW[7] = 0. The background is rendered but the graph overlay is suppressed. This allows the monitor to remain in sync and display an instant image when SW[7] is toggled on.

3. User Controls

3.1 SW[9] — System Power / Reset

Switch Position	System Behavior
SW[9] = 0 (DOWN)	System OFF — sys_reset is HIGH. All synchronous registers clear to zero. Audio driver is reset. LED matrix goes blank. The audio passthrough stops.
SW[9] = 1 (UP)	System ON — sys_reset is LOW. All modules are running. Audio capture, processing, and display are active.

IMPORTANT	SW[9] is the master power switch. Always ensure SW[9] is set to 1 (UP) before expecting any output. When SW[9] = 0, no visual output will appear on the LED matrix regardless of audio input or other switch positions.

3.2 SW[0] — Visualization Mode

Switch Position	Visualization Mode
SW[0] = 0 (DOWN)	Amplitude Time-Domain Mode. Green LED channel active. Scrolling waveform of signal amplitude. Each column = average of 1024 samples (~21ms window). Columns scroll left as time advances.
SW[0] = 1 (UP)	FFT Frequency-Domain Mode. Red LED channel active. Live spectrum analyzer. 16 frequency bins displayed. Bin 0 (leftmost) = lowest frequency (~0–3kHz). Bin 7 (rightmost useful) = ~21kHz. Update rate follows FFT completion.

Live switching: SW[0] can be toggled at any time while the system is running (SW[9] = 1). The display will immediately switch between green amplitude and red frequency visualization without any reset or glitch.

3.3 SW[4:0] — LEDR Debug Bit Selector

Switches SW[4:0] (bits 4 through 0) select which 10-bit window of the 34-bit amplitude accumulator is displayed on the 10 on-board red LEDs (LEDR[9:0]). This is a debug feature for monitoring signal strength.

- LEDR[9:0] = amplitude_accumulator[SW[4:0] +: 10]
- Higher SW[4:0] values observe higher-order (more significant) bits of the accumulator.
- Recommended default: set SW[4:0] = 5–9 to observe mid-range amplitude values.

NOTE	LEDR is a diagnostic tool and is independent of the LED matrix display. It does not affect the visualization output.

3.4 SW[7] — VGA Display Enable

Switch Position	VGA Behavior
SW[7] = 0 (DOWN)	VGA graphs OFF. The module still runs and generates valid sync signals so the monitor stays in sync, but the bar graphs are hidden. Only the background color is shown.
SW[7] = 1 (UP)	VGA graphs ON. Full dual-panel visualization displayed: Amplitude waveform in green on the top half, FFT spectrum in red on the bottom half. Both are shown simultaneously. Independent of SW[0].

NOTE	SW[7] can be toggled at runtime without resetting the system. The VGA_display_wrapper (Apple_Studio_Display_XDR instance) receives the vga_enable signal and applies it combinationally to suppress or enable the graph overlay.

3.5 SW[6] — VGA Background Color

Switch Position	Background Color
SW[6] = 0 (DOWN)	White background. Both the top (Amplitude) and bottom (FFT) panels render on a neutral white canvas. Best for readability under bright lighting conditions.
SW[6] = 1 (UP)	Colored tinted background. The top half of the screen shows a green-tinted background (matching the Amplitude waveform theme). The bottom half shows a red-tinted background (matching the FFT spectrum theme).

NOTE	SW[6] only affects the VGA background rendering. The LED matrix display is unaffected by this switch.

4. Operating the Visualizer

4.1 Quick Start

9. Ensure the LED matrix is connected to GPIO_1.
10. Connect an audio source to the 3.5mm Mic-In port.
11. Optionally connect a VGA monitor to the DE1-SoC VGA port.
12. Program the DE1-SoC with the iWaaave Pro SE bitstream via Quartus Programmer.
13. Flip SW[9] UP (position 1) to power on the system.
14. Set SW[0] DOWN (position 0) for Amplitude Mode or UP (position 1) for FFT Mode.
15. Flip SW[7] UP to enable VGA output. Set SW[6] to choose background color.
16. Play audio from your source. Both the LED matrix and VGA monitor will respond in real time.

4.2 Amplitude Time-Domain Mode (SW[0] = 0)

In this mode, the system continuously captures audio from the left ADC channel, accumulates 1024 samples per window, computes the average absolute magnitude, and maps it to a column height between 0 and 16.

- The display scrolls left in real time: the rightmost column is always the newest sample window, while the leftmost column is the oldest (16 windows back in time).
- A silent input will produce a blank or minimal display.
- A loud, sustained tone will produce tall columns, potentially reaching the top of the matrix.
- The column height is derived from the 5 most significant bits of the 34-bit accumulator after division (bits [33:29]).
- Update rate: approximately 46 frames per second (1024 samples ÷ 48000 Hz ≈ 21ms per frame).

4.3 FFT Frequency-Domain Mode (SW[0] = 1)

In this mode, the system runs a 16-point FFT on incoming audio to decompose the signal into frequency bins. Each bin drives one column of the LED matrix, with the column height representing signal magnitude at that frequency.

Column (left→right)	Frequency Range (approx.)
Column 0	DC / 0 Hz (usually minimal)
Column 1	~0–3 kHz (bass / low-mid)
Column 2	~3–6 kHz
Column 3	~6–9 kHz
Column 4	~9–12 kHz
Column 5	~12–15 kHz
Column 6	~15–18 kHz
Column 7	~18–21 kHz (high frequency)
Columns 8–15	Mirror of bins 0–7 (conjugate symmetric — inherent in real-input FFT)

NOTE	The FFT operates on a 16-sample window with a frequency resolution of 3 kHz per bin (48kHz ÷ 16 = 3kHz). For a coarser but still useful spectrum view, columns 1–7 are the musically relevant bins for typical audio content.

4.4 VGA Display (SW[7] = 1)

When SW[7] is set to 1, the VGA_display_wrapper module drives the external monitor with a fixed dual-panel layout. The top half of the screen always shows the Amplitude waveform in green, and the bottom half always shows the FFT spectrum in red. Both are rendered simultaneously and continuously regardless of what SW[0] is set to.

- Top half (green bars): Amplitude time-domain waveform. Column heights come from waveform_history, the same 16-element shift register that feeds the LED matrix in Amplitude Mode.
- Bottom half (red bars): FFT frequency-domain spectrum. Column heights come from fft_column_heights, the same data that feeds the LED matrix in FFT Mode.
- Background color is determined by SW[6]: white (SW[6]=0) or green-tinted top / red-tinted bottom (SW[6]=1).
- When SW[7] = 0, the background is still rendered but the bar graph bars are hidden.

NOTE	SW[0] controls only the LED matrix (green amplitude or red FFT bars). The VGA display is completely independent — it always shows both panels at the same time regardless of SW[0].

5. Signal Chain Description

This section describes the complete path an audio signal takes through the system, from physical input to LED display and VGA output.

5.1 Audio Capture

audio_driver (audio_driver_BTTBD): The Wolfson WM8731 codec on the DE1-SoC is managed by the audio_driver module, which communicates with the codec over I²C (FPGA_I2C_SCLK / FPGA_I2C_SDAT) and transfers audio data over the serial audio interface (AUD_BCLK, AUD_ADCLRCK, AUD_ADCDAT). On each CLOCK_50 cycle where audio_advance is asserted, a fresh stereo 24-bit sample is available on adc_left and adc_right.

5.2 Absolute Value Computation

Absolute Value Logic: Since audio is a bipolar signed signal (values range from −8388608 to +8388607 in 24-bit two's complement), the system computes the absolute magnitude using combinational logic: abs_audio_left = adc_left[23] ? −adc_left[22:0] : adc_left[22:0]. The sign bit (bit 23) acts as a selector. The result is a 23-bit unsigned magnitude used by the Amplitude processing path.

5.3 Amplitude Processing Path

Sample Accumulator: On every audio_advance pulse, the absolute magnitude is added to a 34-bit accumulator. After 1024 samples, the accumulator holds the sum of 1024 amplitudes. The 5 most significant bits (bits [33:29]) are extracted as new_col_height (range 0–16), and an update_display_signal pulse is generated. The accumulator and counter then reset for the next window.

Waveform Shift Register: When update_display_signal fires, the 16-element array waveform_history shifts left by one position, and new_col_height is inserted at index 15 (the rightmost column). The display always shows the 16 most recent windows, oldest on the left, newest on the right.

5.4 FFT Processing Path

FFT_sample_buffer (uncle_sams_buffer): Collects 16 consecutive 24-bit audio samples and converts them to Q1.14 fixed-point format. When 16 samples are ready, the samples_ready pulse triggers the FFT engine.

FFT_16pt_engine (gentlemen_start_ur_FFT_engines): Performs a 16-point Cooley-Tukey FFT in fixed-point arithmetic. Produces fft_out_re[16] (real parts) and fft_out_im[16] (imaginary parts), asserting done_signal when complete.

FFT_mag_col_h_cal: Computes the magnitude of each bin as $\sqrt{(re^2 + im^2)}$, applies logarithmic scaling, and maps the result to a 5-bit column height (0–16). Outputs fft_column_heights[16] and asserts fft_valid for one clock cycle.

5.5 Display Pipeline

Pixel Mapper + Mode MUX (always_comb): Combinationally translates column height arrays into 16×16 pixel arrays. For each column col and row row, if row < column_height, the corresponding pixel is set to ON. Green pixels (GrnPixels) are filled from waveform_history when SW[0]=0; Red pixels (RedPixels) are filled from fft_column_heights when SW[0]=1.

clock_divider: Produces a scan clock of approximately 1526 Hz by routing bit 14 of the 32-bit ripple counter (50 MHz ÷ 2^{15} = 1525.9 Hz). This scan clock drives the LEDDriver multiplexing loop.

LEDDriver (samsung_AMOLED_driver): Converts RedPixels and GrnPixels into multiplexed row-column signals on the GPIO_1[35:0] header, controlling the 16×16 LED matrix panel in real time.

5.6 VGA Display Pipeline

VGA_display_wrapper (Apple_Studio_Display_XDR): This module receives waveform_history (amplitude column heights) and fft_column_heights directly from the processing pipeline, along with the vga_enable control signal from SW[7] and bg_color_select from SW[6]. It generates standard 640×480 VGA timing signals and drives the 24-bit color data bus to the DE1-SoC on-board VGA DAC.

The screen is permanently divided into two equal horizontal panels. The top 240 lines always display the Amplitude waveform as green vertical bars (sourced from waveform_history). The bottom 240 lines always display the FFT spectrum as red vertical bars (sourced from fft_column_heights). Both panels are rendered simultaneously at all times — this is independent of SW[0], which only controls the LED matrix.

When SW[7] = 0 (vga_enable LOW), the bar graphs are hidden and only the background color is shown, but sync signals remain active so the monitor stays locked. When SW[7] = 1, the full dual-panel visualization appears. SW[6] selects the background: white when LOW, or green-tinted (top) / red-tinted (bottom) when HIGH.

NOTE	The VGA display always shows BOTH the Amplitude waveform (top, green) and the FFT spectrum (bottom, red) at the same time. SW[0] only switches the LED matrix between modes and has no effect on the VGA output.

6. Troubleshooting

Symptom	Likely Cause & Fix
LED matrix is completely blank	SW[9] = 0. Flip SW[9] UP. Verify the bitstream is programmed and CLOCK_50 is running.
No response to audio input	Check that 3.5mm cable is fully inserted in Mic-In. Ensure audio source is active and playing. Verify SW[9] = 1.
Display is always at maximum height	Input signal too loud (ADC saturation). Reduce source volume.
Display shows no response (low amplitude)	Input signal too quiet. Increase source volume, or adjust SW[4:0] to observe higher-order bits on LEDR.
Amplitude mode shows nothing in FFT mode	Verify SW[0] = 1 (UP) for FFT mode. Red pixels only illuminate in FFT mode.
LED matrix flickering or incorrect colors	Verify GPIO_1 connection. Check LEDDriver clock: ensure clk_div[14] is selected (bit 14 of 32-bit counter).
LEDR shows no activity	Set SW[4:0] to bits 5–8 to observe mid-range accumulator bits. SW[4:0]=0 monitors very low-order bits which change too fast to see on LEDs.
FFT shows only DC component	No audio input signal. Ensure audio is actively playing into Mic-In.
VGA monitor shows no image	Check VGA cable connection. Verify SW[9]=1 (system on). Flip SW[7] to 1 (UP) to enable VGA output.
VGA shows background but no graphs	SW[7]=0 (VGA graphs disabled). Flip SW[7] UP to enable graph overlay.
VGA image is solid color / wrong color	Check SW[6] position: SW[6]=0 = white bg, SW[6]=1 = green/red tinted bg.
VGA shows only LED mode, not both panels	VGA always shows both amplitude (top) and FFT (bottom) simultaneously. SW[0] only affects the LED matrix output. Verify SW[7]=1 is set.

7. Technical Specifications

Parameter	Value
FPGA	Intel Cyclone V (DE1-SoC)
System Clock	50 MHz (CLOCK_50)
Audio Codec	Wolfson WM8731 (on-board)
Sample Rate	48,000 Hz (48 kHz)
Bit Depth	24-bit signed (two's complement)
Amplitude Window	1024 samples per update window
Amplitude Update Rate	≈21.33 ms / frame (≅46.9 FPS)

Parameter	Value
FFT Points	16-point Cooley-Tukey FFT
FFT Freq Resolution	3000 Hz / bin (48kHz ÷ 16)
Accumulator Width	34-bit (handles up to 1024 × 2^23)
Column Height Range	0 to 16 (5-bit unsigned)
LED Matrix	16×16, Red + Green channels, GPIO_1[35:0]
LED Scan Clock	≈1526 Hz (CLOCK_50 ÷ 2^15)
Audio Input	3.5mm TRS Mic-In (AUD_ADCDAT, AUD_BCLK, AUD_ADCLRCK, AUD_DACLRCK)
Audio Output (passthrough)	DAC output (AUD_DACDAT, AUD_DACLRCK)
I²C Codec Control	FPGA_I2C_SCLK / FPGA_I2C_SDAT (inout wire)
VGA Resolution	640×480 pixels (standard VGA)
VGA Pixel Clock	≈25.175 MHz (generated from CLOCK_50)
VGA Color Depth	24-bit RGB (8-bit per channel via DE1-SoC VGA DAC)
VGA Connector	15-pin D-Sub on DE1-SoC board
VGA Top Panel	Amplitude waveform — green bars (waveform_history[0:15])
VGA Bottom Panel	FFT spectrum — red bars (fft_column_heights[0:15])
VGA Signals	VGA_R[7:0], VGA_G[7:0], VGA_B[7:0], VGA_HS, VGA_VS, VGA_CLK, VGA_BLANK_N, VGA_SYNC_N
User Switches	SW[9]: sys_reset · SW[0]: LED mode · SW[7]: VGA enable · SW[6]: VGA bg color · SW[4:0]: LEDR debug
Debug Outputs	LEDR[9:0] on-board LEDs

8. Developer Notes

8.1 FPGA_I2C_SDAT Wire Type

The I²C data line FPGA_I2C_SDAT is declared as inout wire, not logic. This is required because inout (bidirectional) ports in Verilog must be connected to explicit wire-type signals — Quartus cannot infer bidirectional driver control from logic type. The audio_driver module manages the tristate control internally.

8.2 Signed Audio Samples in FFT

The FFT_top module receives the raw signed 24-bit sample adc_left (not the absolute value). The FFT requires the actual bipolar signal to correctly compute both positive and negative frequency components. The absolute value (abs_audio_left) is used only in the Amplitude processing path.

8.3 Extending to 256-Point FFT

The design includes commented-out support for a 256-point FFT engine (FFT_256pt_engine) and corresponding 256-element sample buffer. Switching to 256 points would increase frequency resolution to 48000 ÷ 256 ≈ 188 Hz/bin at the cost of higher latency and resource utilization.

8.4 Color Extension

The Pixel Mapper only populates GrnPixels or RedPixels at a time, leaving the unused channel at zero. Both channels can be driven simultaneously by extending the always_comb block to merge data from both processing paths, enabling yellow (R+G) as a third color.

8.5 Increment Notation

Developer note from source code: In the always_ff accumulator block, sample_collected_count is incremented with + 1'b1 for explicit 1-bit literal typing. The expressions + '1 and + 1 also work in SystemVerilog but may require care with bit-width context, especially in synthesis tools.

8.6 VGA Dual-Panel Architecture

The VGA_display_wrapper (Apple_Studio_Display_XDR) receives waveform_history and fft_column_heights directly from the top-level module, bypassing the LED Pixel Mapper entirely. The VGA layout is fixed: the top half of the 640×480 frame always renders the Amplitude waveform as green bars, and the bottom half always renders the FFT spectrum as red bars. Both are shown simultaneously at all times. This is architecturally distinct from the LED matrix, where SW[0] selects a single mode. The VGA always shows both. Column widths are computed as 640 ÷ 16 = 40 pixels per bin; bar heights scale proportionally to the panel height (240px).

8.7 VGA Background Color Control (SW[6])

When SW[6] = 0, both panel backgrounds are white. When SW[6] = 1, the background of the top Amplitude panel is tinted green and the bottom FFT panel is tinted red, creating a visual distinction that reinforces the color theme of each visualization. The bar graph colors themselves (green for amplitude, red for FFT) remain constant regardless of SW[6].

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